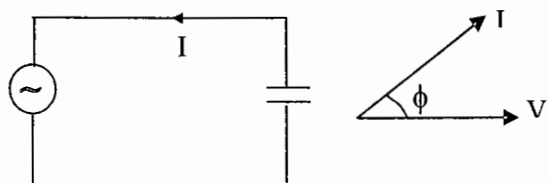




The reactive power absorbs by the Inductor is positive. The reactive power of synchronous machine will also positive and it will act as the synchronous capacitor.

Case (2): The Synchronous Machine is connected in parallel with shunt capacitor.



The shunt capacitor will deliver the required reactive power to the synchronous machine. The reactive power absorbed by shunt capacitor is negative. That is capacitor delivers reactive power and it will be absorbed by synchronous machine so it act as synchronous Inductor.

Electrical equipments - Reactive power nature:

Induction motor – absorption (sink)

Transformer – absorption (sink)

Transmission line – Neither sink nor source.

(loading = SIL)

- Sink (loading > SIL)

- Source (loading < SIL)

Under ground cable:- Source

Synchronous generator

- Over excitation – source of Q (lagging pf)
- Under excitation – sink of Q (leading pf)
- Normal excitation – neither sink nor source of Q (unity pf)

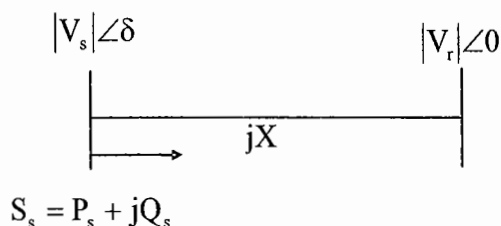
Synchronous Motor

- Over excitation – source of Q (leading pf)
- Under excitation – sink of Q (lagging pf)
- Normal excitation – neither sink nor source of Q (unity pf)

Any no loaded over excitation synchronous machine will act as synchronous capacitor and no-

loaded under excitation synchronous machine will act as synchronous Inductor.

Relation between receiving and voltage $|V_r|$ & net reactive power (Q_{net} (or) Q_s) :



So just we have to find as the we get relation between Q_s & $|V_r|$

$$S_s = V_s I_s^*$$

$$S_s = |V_s| \angle \delta \left(\frac{|V_s| \angle \delta - |V_r| \angle 0}{jX} \right)$$

$$= |V_s| \angle \delta \left(\frac{|V_s|}{X} \angle \delta - 90^\circ - \frac{|V_r|}{X} \angle -90^\circ \right)^*$$

$$= |V_s| \angle \delta \left(\frac{|V_s|}{X} \angle 90^\circ - \delta - \frac{|V_r|}{X} \angle 90^\circ \right)$$

$$S_s = \frac{|V_s|}{X} \angle 90^\circ - \frac{|V_s||V_r|}{X} \angle 90^\circ + \delta$$

Reactive power

$$Q_s = \frac{|V_s|^2}{X} \sin 90^\circ - \frac{|V_s||V_r|}{X} \sin(90 + \delta)$$

$$Q_s = \frac{|V_s|}{X} (|V_s| - |V_r| \cos \delta)$$

Assuming that the system is more stable $\cos \delta$ approximately equal to 1 .

$$Q_s = \frac{|V_s|}{X} (|V_s| - |V_r|)$$

The magnitude of receiving end voltage

$$|V_r| = |V_s| - \frac{X}{|V_s|} Q_s$$

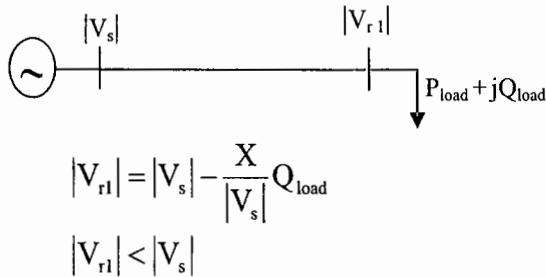
Where Q_s is nothing but net reactive power in the system which will be modified in order to change magnitude of receiving end voltage.

Shunt Capacitor:

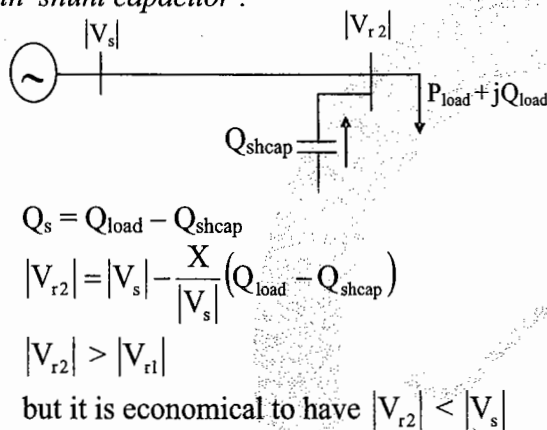
- It will modify current carried by the line

- It will control under voltage in the system
- It will inject reactive power into the system

without shunt capacitor :

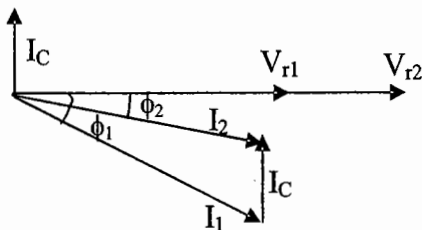


With shunt capacitor :

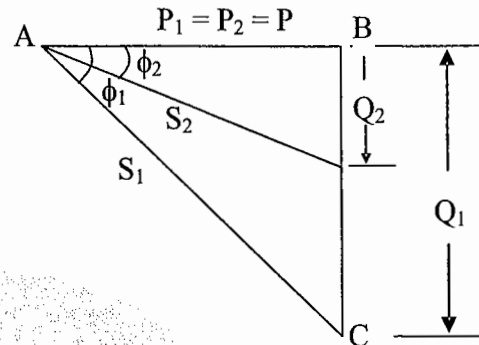


Whenever $|V_{r2}| = |V_s|$, the reactive power required is very high so that the rating of the shunt capacitor will increase which in then increase the cost of supply. The shunt capacitor will also improve the power factor of the load apart from improvement in voltage. However the improvement in power factor is high when compared to improvement in voltage.

So practically shunt capacitor will be treated as power factor correction device .



The improved power factor will reduce the KVA demand of the consumer so that the fixed tariff will reduced. Due to shunt capacitor there is no change in real power demand.



$$\phi_2 < \phi_1$$

$$\cos \phi_2 > \cos \phi_1$$

The reactive power supplied by the shunt Capacitor = $Q_1 - Q_2$

$$Q_{shcap} = S_1 \sin \phi_1 - S_2 \sin \phi_2$$

$$Q_{shcap} = P (\tan \phi_1 - \tan \phi_2)$$

If P is 3-phase real power consumed by the load then Q_{shcap} is the reactive power require to be injected by capacitor bank .

During operation,

$$Q_{shcap} / ph = \frac{V_{ph}^2}{X_c / ph}$$

$$Q_{shcap} (3 - \phi) = 3 \frac{V_{ph}^2}{X_c / ph}$$

$$Q_{shcap} (3 - \phi) = 3 V_{ph}^2 \omega C / ph$$

$$Q_{shcap} \propto V^2 f$$

At the designing stage ,

$$C / ph = \frac{Q_{shcap} (3 - \phi)}{3 V_{ph}^2 \omega}$$

Normally the shunt capacitors are connected in delta configuration across the supply.



Example 2.6.1:

At an industrial sub – station with a 4 mw load, a capacitor of 2 MVAR is installed to maintain the load power factor. at 0.97 lag. If the capacitor bank is out of service, the load power factor will be

- (a) 0.8 lag (b) 0.75 lag
(c) 0.9 lag (d) 0.85 lag

Sol: Reactive power supplied = $P[\tan\phi_1 - \tan\phi_2]$

$$2 = 4 [\tan\phi_1 - \tan(\cos^{-1}(0.97))]$$

$$\tan\phi_1 = 0.7506$$

$$\phi_1 = \tan^{-1}(0.7506) = 36.86$$

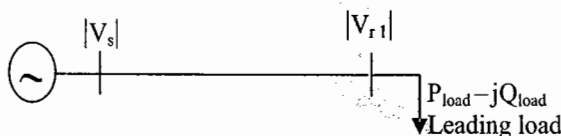
$$\text{Power factor} = \cos\phi$$

$$= \cos(36.86) = 0.8 \text{ lag}$$

Shunt Inductor :

When the transmission line is operating at no-load (or) light load, the receiving end voltage is more than the sending end voltage. In order to reduce the receiving end voltage a suitable shunt inductor is connected at the load bus to absorb excess reactive power from the system.

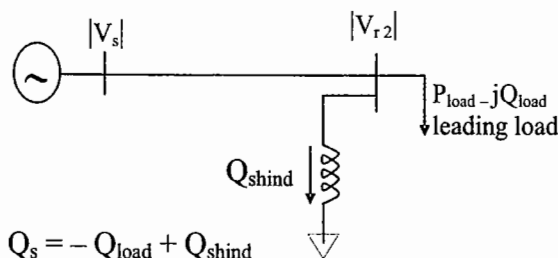
without shunt inductor :



$$|V_{r1}| = |V_s| + \frac{X}{|V_s|} Q_{\text{load}}$$

$$|V_{r1}| > |V_s|$$

With shunt inductor :

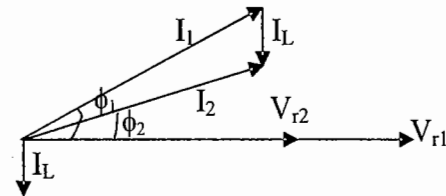


$$|V_{r2}| = |V_s| - \frac{X}{|V_s|} (-Q_{\text{load}} + Q_{\text{shind}})$$

$$|V_{r2}| = |V_s| + \frac{X}{|V_s|} (Q_{\text{load}} - Q_{\text{shind}})$$

$$|V_{r2}| < |V_{r1}|$$

but it is economical to have $|V_{r2}| > |V_s|$



$$\phi_2 < \phi_1$$

$$\cos\phi_2 > \cos\phi_1$$

Power factor will be improved by placing shunt reactor only when ever the load is leading which is not practical so shunt reactor is not a practical power factor correction device .

Practical shunt reactor will be called as Ferranti effect control device.

The reactive power absorbed by shunt Inductor

$$Q_{\text{shind}} = P (\tan\phi_2 - \tan\phi_1)$$

During operation

$$Q_{\text{shind}} / \text{ph} = \frac{V_{\text{ph}}^2}{X_L / \text{ph}}$$

$$Q_{\text{shind}} (3 - \phi) = 3 \frac{V_{\text{ph}}^2}{X_L / \text{ph}}$$

$$Q_{\text{shind}} (3 - \phi) = 3 \frac{V_{\text{ph}}^2}{\omega L / \text{ph}}$$

$$Q_{\text{shind}} \propto \frac{V^2}{f}$$

At the designing stage ,

$$L / \text{ph} = \frac{3V_{\text{ph}}^2}{\omega Q_{\text{shind}} (3 - \phi)}$$



Example 2.6.2:

In a 400 kV power network, 360 kV is recorded at a 400 kV bus. The reactive power absorbed by a shunt reactor rated for 50 MVAR, 400 kV connected at the bus is _____

Sol: At rated voltage (400 kV) $Q_{\text{reactor}} = 50 \text{ MVAR}$

$$Q_{\text{reactor}} \propto V^2$$

At $V_1 = 400 \text{ kV}$, $Q_{\text{reactor}1} = 50 \text{ MVAR}$

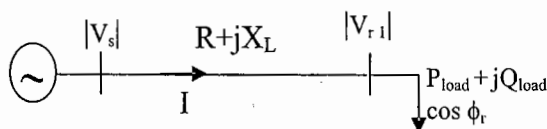
At $V_2 = 360 \text{ kV}$,

$$\begin{aligned} Q_{\text{reactor}2} &= Q_{\text{reactor}1} \times \frac{V_2^2}{V_1^2} \\ &= 50 \times \frac{360^2}{400^2} \\ &= 40.5 \text{ MVAR} \end{aligned}$$

Series Capacitor:

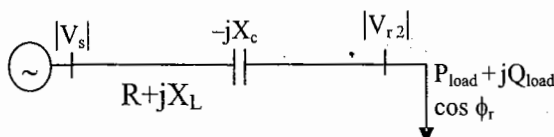
The voltage control can be done by changing reactance of transmission line.

without series capacitor :



$$\Delta V_1 = IR \cos \phi_r + IX_L \sin \phi_r$$

With series capacitor :



$$\Delta V_2 = IR \cos \phi_r + I(X_L - X_C) \sin \phi_r$$

$$\Delta V_2 < \Delta V_1, |V_{r2}| > |V_{r1}|$$

The series compensation will be considered if (X/R) is very high

The percentage compensation is $\frac{X_C}{X_L} \times 100$

The amount of reactive power of series capacitor will be $Q_{\text{secap}}/\text{ph} = I^2 X_C / \text{ph}$

The voltage boost produced by series capacitor

$$\Delta V_C = I X_C \sin \phi_r$$

Due to series capacitor

- Maximum power transfer capability and stability of system increases.
- Fault level in the system increases
- During fault a large voltage appears across series capacitor such that the capacitor may be damaged.

So the series capacitor will be protected by using spark gap.

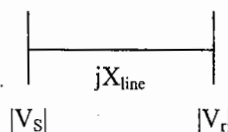
- There is a subsynchronous resonance occurs in the system such that shaft of alternator will be damaged.

Example 2.6.3:

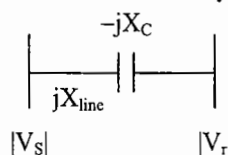
The steady state stability limit of a transmission line is 400 MW. If 30% series compensation is added to the line then the new SSSL of transmission line will be (Resistance & capacitance of line are neglected, voltages at both ends of line are remains unchanged)

- (a) 400 MW (b) 571.4 MW
(c) 280 MW (d) 1333.3 MW

Sol: $P_{\text{max}} = \frac{|V_s| |V_r|}{X_{\text{line}}} = 400 \text{ MW}$



With series compensation



$$P_{\text{max (new)}} = \frac{|V_s| |V_r|}{X_{\text{line}} - X_C}$$

$$\% \text{ series compensation} = \frac{X_C}{X_{\text{line}}} \times 100\%$$

$$\frac{X_C}{X_{\text{line}}} = 0.3$$

$$X_C = 0.3 X_{\text{line}}$$



$$\begin{aligned}\therefore P_{\max(\text{new})} &= \frac{|V_s| |V_r|}{0.7 X_{\text{line}}} \\ &= \frac{400}{0.7} \text{ MW} = 571.5 \text{ MW}\end{aligned}$$

Example 2.6.4:

If a line is 100% series compensated it may result into series resonance at

- (a) power frequency (50Hz) (b) 100 Hz
(c) 150 Hz (d) none

Ans: (a)

Differences between shunt and series capacitors:

Shunt Capacitor:

- The voltage produced by shunt capacitor is more uniformly distributed throughout its length.
- There is no change in stability.
- There is an improvement in load power factor.
- For the same voltage boost, the reactive power required is high.

Series Capacitor:

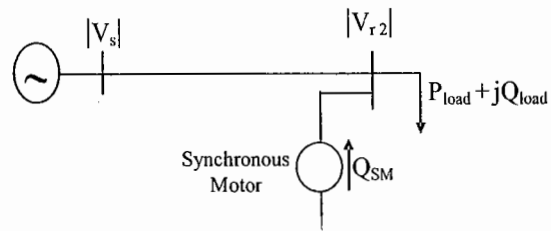
- The voltage produced by series capacitor is sudden and boost at the points where it is connected.
- It will improve the stability.
- No improvement in load power factor.
- For the same voltage boost, the reactive power required is less.

Synchronous Condensor or capacitor:

It is an over excited synchronous motor operating at no load condition .

At no load condition real power drawn by the synchronous motor is zero where as the reactive power injected by the synchronous motor into the bus given as

$$Q_{\text{SM}}(\text{Supply}) = \frac{|V|}{X_s} (|E| - |V|)$$



During over excitation field current given to rotor field winding is very high so the flux produced is more such that $|E| > |V|$.

The synchronous condensor is similar to shunt capacitor so this can also be called as power factor correction device .

The rating of synchronous condensor for power factor correction is

$$Q_{\text{syncond}} = P (\tan \phi_1 - \tan \phi_2)$$

Power factor of synchronous condenser is ZPF lead.

Synchronous Coil or Inductor :

It is an under excited synchronous motor operating at no load condition .

At no load condition real power drawn by the synchronous motor is zero where as the reactive power will be absorbed by the synchronous motor from the bus which is given as

$$Q_{\text{SM}}(\text{abs}) = \frac{|V|}{X_s} (|V| - |E|)$$

During under excitation field current given to rotor field winding is less so the flux produced is low such that $|E| < |V|$.

The synchronous coil is similar to shunt inductor so this can also be called as Ferranti effect control device .

Power factor of synchronous coil is ZPF lag.

Differences between shunt capacitor and synchronous condenser:

Shunt Capacitor:

- (1) There is a step control in reactive power.
- (2) The cost is less.
- (3) There is no consumption of real power.

- (4) There is no stability problem even large variation at load voltages.
- (5) There is no effect on fault currents.

Synchronous Capacitor:

- (1) There is smooth control over the reactive power.
- (2) The cost is high
- (3) Due to mechanical loss, there is some real power loss even though it is under no-load.
- (4) The synchronous motor will get asynchronised if there is large variation in voltages.
- (5) It will rise the fault levels during short circuit fault.

Example 2.6.5:

An induction Motor operating at 0.8 p.f lag consuming 300 kw. A zero real power consuming synchronous motor is connected across the induction motor to raise the p.f. to 0.92 lagging. The reactive power drawn by the synchronous motor is

- (a) 97.2 kVAr lag (b) 97.2 kVAr lead
(c) 36 kVAr lag (d) 36 kVAr lead

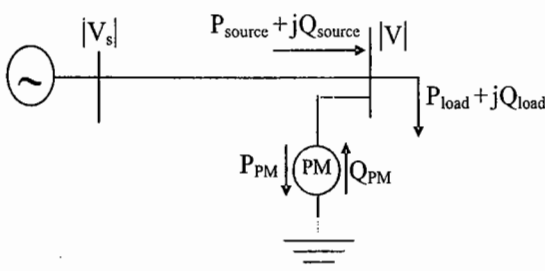
Sol: Reactive power supplied = $P(\tan\phi_1 - \tan\phi_2)$
 $= 300[\tan(\cos^{-1} 0.8) - \tan(\cos^{-1} 0.92)]$
 $= 97.2 \text{ kVAr}$

Lagging reactive power supplied means leading reactive power absorption

$\therefore$ Synchronous motor will draw 97.2 kVAR lead.

Synchronous Phase Modifier:

The loaded synchronous motor with over excitation is called synchronous phase modifier.



$$P_{\text{source}} = P_{\text{load}} + P_{\text{PM}}$$

$$Q_{\text{source}} = Q_{\text{load}} - Q_{\text{PM}}$$

Where

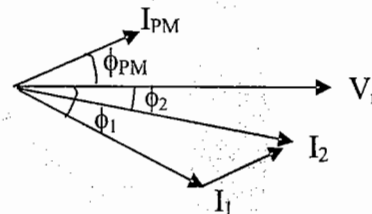
$$P_{PM} = \frac{|E||V|}{X_s} \sin \delta$$

$$Q_{PM} = \frac{|V|}{X_S} [|E| \cos \delta - |V|]$$

The phase modifier will inject reactive power only for $|E|\cos \delta > |V|$

The power factor of phase modifier

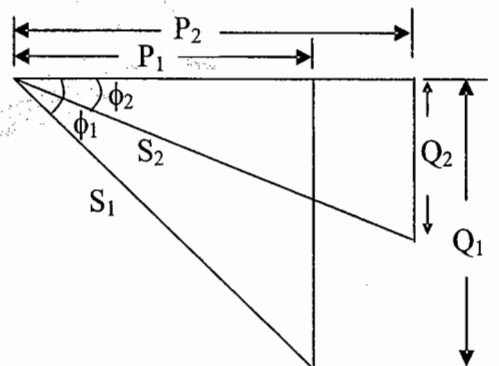
$$\cos \phi_{PM} = \frac{P_{PM}}{\sqrt{P_{PM}^2 + Q_{PM}^2}}$$



$$\phi_2 > \phi_1$$

$$\cos\phi_2 > \cos\phi_1$$

power factor got improved .



The reactive power supplied by synchronous phase modifier

$$= Q_1 - Q_2$$

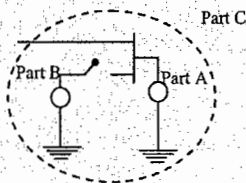
$$= S_1 \sin \phi_1 - S_2 \sin \phi_2$$

$$Q_{PM}(\text{Supply}) = P_1 \tan \phi_1 - P_2 \tan \phi_2$$



Example 2.6.6:

In the following system, before connections part B, part C power factor is 0.6 lag. After connections part B, part C power factor is 0.8 lag. Then which of the following option is correct.



- (a) Part –A is an under excited synchronous motor, part – B is a shunt capacitor
- (b) Part –A is an over excited synchronous motor, part – B is a shunt capacitor
- (c) Part –A is an over excited synchronous motor, part – B is a shunt inductor
- (d) Part –A is an under excited synchronous motor, part – B is a shunt inductor

Ans: (a)

2.7 Under Ground Cables

The combination of conductor and its insulator is called cable.

All electrical cables consists of three essential points:

- i) The conductor for transmitting electrical power,
- ii) The insulation, an electrical insulating medium, needed to insulate the conductor from direct contact with earth or other object.
- iii) External protection against mechanical damage, chemical or electro chemical attacks, fire or any other dangerous effects external to the cables.
 - Copper conductor has extensively been used for cable but of aluminium is being used to a considerable extent.
 - A stranded conductor is expressed as $\frac{n}{d}$ where the first number stands for the number of strands used and the second number corresponds to the diameter of each strand in mm. Sometimes the second number given corresponds to the gauge of the strand used e.g., 3/20 which means a cable with three strands each of 20 SWG.

The main requirements of the insulating materials used for cable are:

- i) high insulation resistance
- ii) high dielectric strength
- iii) good mechanical properties

Various insulations:

- i) Vulcanized rubber
Dielectric strength is 10 kV/mm to 20kV/mm.
- ii) Butyl rubber
- iii) Silicon rubber
Dielectric strength is 1.2 kV/mm to 3kV/mm.
- iv) Neoprene rubber
- v) Polyvinyl chloride
Dielectric strength is 17 kV/mm.
- vi) Impregnated paper.
Dielectric strength is 20 kV/mm to 30kV/mm.

Under Ground Cables are used in the place of overhead lines to have following advantages and disadvantages:

Advantages:

- 1. In a densely populated circuits where overhead lines are not possible.
- 2. Under ground cables provides better regulation
- 3. The chances of accidents in under ground system are very low compared to over head lines.
- 4. As the cables are laid under ground with better insulation, the chances of failure or fault are less compared to over head lines.



Disadvantages:

1. Under ground cables are very costly as compared to overhead lines.
2. Practically, identification of cable faults is difficult than in case of faults in the overhead lines.
3. Jointing of cables is difficult. Hence tapping for loads and service mains is not convenient in under ground system.

Classification of Cables:

1. According to number of conductors.

It can be classified as single, two, three or four core cables.

2. According to the voltage

- i) Low – tension (L.T) cables upto 1000
- ii) High – tension (H.T) cables upto 11,000
- iii) Super – tension (S.T) cables upto 22kV to 33kV
- iv) Extra high – tension (E.H.T) cables from 33kV to 66kV
- v) Extra super voltage (E.S.V) cables – beyond 132kV.

3. According to the nature of the application of insulation and lead sheath.

- i) Belted type ii) H- type
- iii) S. L. type iv) H. S. L.

4. According to the methods of improving the dielectric stress

- i) Solid type upto 33 kV
- ii) Oil filled type upto 500kV
- iii) Gas pressure type. upto 235kV

Construction of Cables:

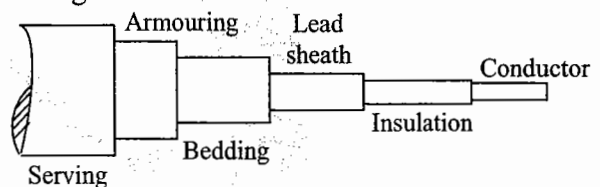
- a) **Cores or conductors:** The conductors are made of tinned, copper or aluminum and are usually stranded in order to provide flexibility to the cable.
- b) **Insulation:** - Each conductor is provided with a suitable thickness of insulation, depending upon the voltage the commonly used materials for insulation are impregnated paper, varnished cambric or rubber mineral compound.

c) **Metallic Sheath:** - It is provided to avoid the entry of moisture content from earth surface to insulator. The material used for sheath is Al (or) lead.

d) **Bedding:** It consists of a fibrous material like jute or hessian. The purpose of bedding is to protect the metal sheath against corrosion and from mechanical injury due to Armouring.

d) **Armouring:** It consists of one or two layers of galvanized steel wire or steel tape. Its purpose is to protect the cable from mechanical injury while laying it and during the course of handling.

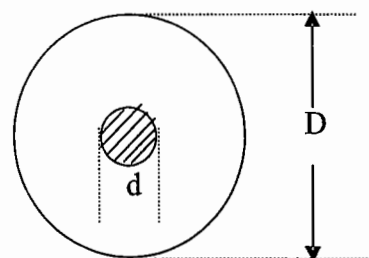
e) **Serving:** - In order to protect Armouring from atmospheric conditions, a layer of fibrous material (like jute) similar to bedding is provided over the Armouring this is known as serving.



Types of cables:

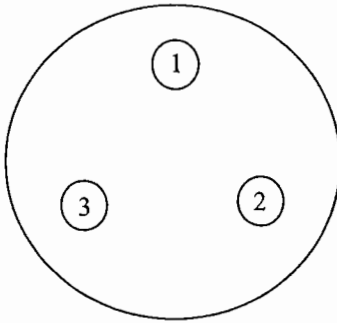
- 1) Single core
- 2) Three core.

1. **Single core:** It consists of stranded copper conductor a belt of insulation of impregnated paper and lead sheath over it. The sheath is protected by covering it with hessian tapes or jute which is soaked in some preservative compound of bituminous nature.



2. Three core cable:

This cable is used for the 3 phase power transmission in which all the three cores will be placed inside a single cable.



Insulation resistance of single core cable:

Let ' ℓ ' be the length of the cable in meter.

' ρ ' be the resistivity of insulator in Ω -meter

' r ' be the radius of single core cable of conductor.

' R ' be the internal sheath radius

$$\text{Insulation resistance, } R_i = \frac{\rho}{2\pi\ell} \ln\left(\frac{R}{r}\right)$$

The insulation resistance of a cable is inversely proportional to its length.

Capacitance of a single core cable :

Let ' R ' be the internal sheath radius

' r ' be the conductor radius

$$\epsilon = \epsilon_0 \epsilon_r$$

ϵ_0 is the permittivity of free space

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

' ϵ_r ' is the relative permittivity of the insulation.

' λ ' be the charge per meter axial length of the cable in coulombs.

Potential difference between the conductor and sheath.

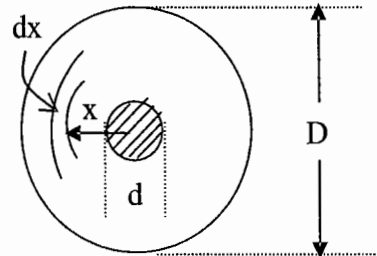
$$V = \frac{\lambda}{2\pi\epsilon} \ln\left(\frac{R}{r}\right)$$

Capacitance of the cable is $C = \frac{\lambda}{V}$

$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln\left(\frac{R}{r}\right)} \text{ F/m}$$

Electro static stress distribution in cable cross section:

V = potential difference between core and sheath



Electric field intensity at any point (x) in cable cross section is

$$E_x = \frac{\lambda}{2\pi\epsilon x}$$

$$E_x = \frac{V}{x \ln\left(\frac{R}{r}\right)} \text{ kV/cm}$$

On surface of conductor ($x = r$),

$$E_r = \frac{V}{r \ln\left(\frac{R}{r}\right)} \text{ kV/cm}$$

On surface of sheath ($x = R$),

$$E_R = \frac{V}{R \ln\left(\frac{R}{r}\right)} \text{ kV/cm}$$

The electrical stress is maximum at the surface of conductor and the stress is minimum on the surface of sheath.

By distributing the stress uniformly, the breakdown of insulation can be avoided this may be Two methods.

(a) by using metallic inter sheaths

(b) by using insulating materials of different dielectric constants.

Inter sheath grading: The inter sheaths made of metallic cylinders one or more are inserted in the dielectric between the conductor and lead sheath to fix up the potentials at that distance from the surface of core in the insulation. The inter sheaths do not carry any part of working current, but carries the current which is the difference between

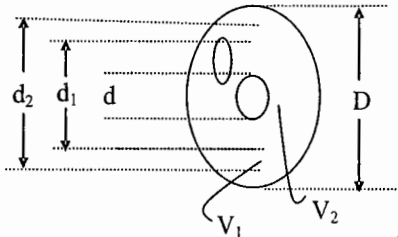
the charging currents taken by section on each side.

3 layers. Two inter sheath are inserted between cable throughout length.

$$g_{1\max} = V - V_1/r \ln(d_1/d)$$

$$g_{2\max} = V_1 - V_2/r_1 \ln(d_2/d)$$

$$g_{3\max} = V_2/r_2 \ln(D/d_2)$$



The stress can be made to vary between the same maximum and minimum value by choosing d_1 and d_2 such that d_1/d

$$= d_2/d_1 = D/d_2 = \alpha$$

$$\alpha^3 = D/d; \quad \alpha = \sqrt[3]{D/d}$$

$$d_1 = \alpha d \quad d_2 = \alpha^2 d$$

the max stress are required to be made equal, we have $g_{1\max} = g_{2\max} = g_{3\max}$

$$V_2 = V_1/(1+1/\alpha) = V/(1+1/\alpha+1/\alpha^2)$$

One intersheath

$$d_1/d = D/d_1 = \alpha$$

$$\alpha^2 = D/d \quad \alpha = \sqrt{D/d}$$

$$d_1 = \alpha d$$

$$V_1 = (1+1/\alpha) V$$

$$(1+1/\alpha+1/\alpha^2)$$

$g_{\max}$ with two intersheath / $g_{\max}$ with out intersheath = $\frac{V}{\frac{1}{3}(1+\alpha+\alpha^2)\frac{d}{2}\ln(D/d)}$

$$= \frac{V}{d/2 \ln(D/d)}$$

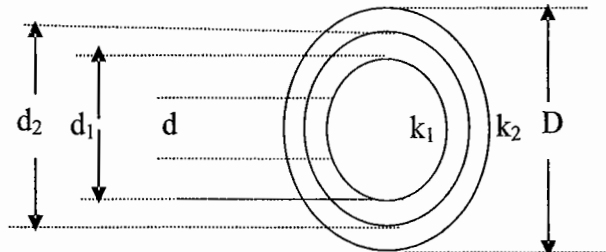
$$= \frac{3}{(1+\alpha+\alpha^2)} < 1.0$$

$g_{\max}$ with one intersheath / $g_{\max}$ with out intersheath = $2/(1+\alpha)$

The new max stress with two intersheath is only $3/(1+\alpha+\alpha^2)$ Times that of stress with out any intersheath.

Capacitance Grading:

The insulation may be made of dielectrics of different permittivity such a cable is known as graded cable and the arrangement results in a more uniform stress in the dielectric.



Let d_1 be the diameter of the dielectric having permittivity k_1 and D the diameter of the dielectric having permittivity k_2 .

$$= \frac{2\pi}{\frac{1}{k_1} \ln(d_1/d) + \frac{1}{k_2} \ln(D/d_1)}$$

$$g_{1\max} = Q/\pi k_1 d \quad \text{'Q' in times of V}$$

$$g_{1\max} = \frac{2V}{d[\ln(d_1/d) + k_1/k_2 \ln(D/d_1)]}$$

Most economical radius of core(r):

The most economical radius of core (r) will be designed in order to minimize 'E' on the surface of core by keeping operating voltage(V) and radius of cable(R) as constant is

$$\frac{R}{r} = e = 2.718$$

$$\frac{r}{R} = \frac{1}{e} = 0.368$$

The most economical radius of core is $r = 0.368R$

Example 2.7.1:

A single-core length 5 km having diameter of core as 2 cm and diameter of cable as 4 cm. if the resistivity of dielectric material is $1.6 \times 10^{12} \Omega\text{-m}$ and $\epsilon_r = 4$ then what is insulation resistance for most economical radius of core.

Sol: With most economical radius of core,

$$R_{\text{ins}} = \frac{\rho_{\text{ins}}}{2\pi\ell}$$

$$= \frac{1.6 \times 10^{12}}{2\pi \times 5000}$$

$$= 50.93 \text{ M}\Omega$$

Power factor in cables:

The capacitance effect of the dielectric and the intersheath, a charging current flows from the supply the value of this charging current being $V/1/\omega C = \omega CV$

Where $\omega = 2\pi f$

The charging current leads the voltage V by 90° and the total current drawn from the supply is the Phasor sum of currents V/R_i and ωCV shown in below fig.

The total current I leads the voltage by an angle ϕ where $\cos \phi$ gives the power factor of the cable.

$$\cos \phi = \cos(\pi/2 - \delta)$$

$$= \sin \delta = \delta \text{ (since } \delta \text{ is small).}$$

In other words power factor of the cable $= \delta$

When δ is small, $\sin \delta = \tan \delta = \delta$

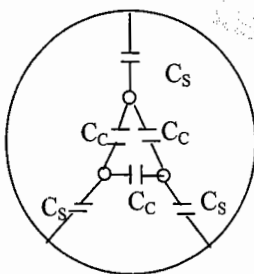
Therefore, $\delta = V/R_i \cdot 1/\omega C V = 1/\omega C R_i$

Where $1/R_i = G = \text{conductance of the cable.}$

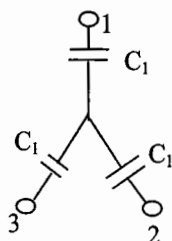
The power factor of the cable is $G/\omega C$.

Capacitance in 3-core cable:

The three-core cable has capacitance between the cores and each core has capacitance with sheath is shown below fig.



Capacitances C_c to the core are in delta and can be replaced by an equivalent star arrangement shown in below fig.



The impedance between core 1 and the star point must be equal to $(1/3)$ times the impedance of each branch of the delta, this gives

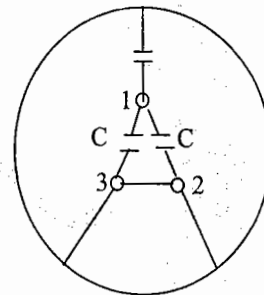
$$1/\omega C_1 = 1/3 \omega C_c \quad (\text{or}) \quad C_1 = 3C_c$$

The star may be assumed to be at zero potential and if sheath is also at zero potential the capacitance of each conductor to neutral is

$$C_0 = C_1 + C_s = 3C_c + C_s$$

Methods of calculation of C_s & C_c :

Method 1: Let conductors 2 and 3 be connected to the sheath. Capacities C_c between conductors 2 and 3 and C_s of conductors 2 and 3 with respect to the sheath are eliminated.



Capacitances C_c and C_s are now in parallel across core one and the sheath and they add up. Measure the capacitance between core one and the sheath, is

$$C_1 = C_s + 2C_c \dots \dots \dots (1)$$

Method 2: All the conductors are connected together and capacitance C_2 is measured between them and sheath.

$$C_2 = 3C_s \quad \text{or}$$

$$C_s = 1/3 C_2 \dots \dots \dots (2)$$

$$\text{Since } 2C_c = C_1 - C_s$$

$$C_c = 1/2 (C_1 - 1/3 C_2)$$

$$= 1/6 (3C_1 - C_2)$$

$$C_n = C \text{ per phase}$$

$$\text{From 1 and 2} \quad C_n = C_s + 3C_c = 3/2 C_1 - C_2/6$$

